

Current and Future Techniques in the Training of Large Reasoning Models

Gavin Lynch¹

1. Department of Computer Science and Software Engineering, California Polytechnic State University

Abstract

Large Reasoning Models (LRMs) are a new class of AI systems designed to enhance logical reasoning, multi-step inference, and structured problem-solving, distinguishing them from traditional AI models and Large Language Models (LLMs). This paper explores key examples, including OpenAI's o1, DeepSeek's R1, and others, highlighting their unique training methodologies. Unlike LLMs, LRMs leverage structured reasoning datasets, Reinforcement Learning with Human Feedback (RLHF), Chain-of-Thought (CoT) prompting, and iterative problem-solving. Deep-Seek's approach, incorporating cold-start training and reasoning-oriented reinforcement learning, demonstrates alternative strategies for enhancing logical reasoning. Additionally, the paper examines AI-generated training data, rejection sampling, and ethical concerns surrounding synthetic reasoning datasets. This paper also explores emerging methodologies in LRM development, anticipating advancements such as including knowledge graphs, integration with external databases, and retrieval-augmented generation (RAG). These innovations are expected to enhance reasoning capabilities, enabling more sophisticated decision-making and automation across various domains. This paper is meant to explore the current landscape of LRMs and give insight into potential future research directions.

Keywords

artificial intelligence, computing and processing, machine learning

Current and Future Techniques in the Training of Large Reasoning Models

Gavin Lynch¹

¹Department of Computer Science and Software Engineering, California Polytechnic State University, San Luis Obispo, CA 93407, USA

March 12, 2025

Abstract

Large Reasoning Models (LRMs) are a new class of AI systems designed to enhance logical reasoning, multi-step inference, and structured problem-solving, distinguishing them from traditional AI models and Large Language Models (LLMs). This paper explores key examples, including OpenAI’s o1, DeepSeek’s R1, and others, highlighting their unique training methodologies. Unlike LLMs, LRMs leverage structured reasoning datasets, Reinforcement Learning with Human Feedback (RLHF), Chain-of-Thought (CoT) prompting, and iterative problem-solving. DeepSeek’s approach, incorporating cold-start training and reasoning-oriented reinforcement learning, demonstrates alternative strategies for enhancing logical reasoning. Additionally, the paper examines AI-generated training data, rejection sampling, and ethical concerns surrounding synthetic reasoning datasets. This paper also explores emerging methodologies in LRM development, anticipating advancements such as including knowledge graphs, integration with external databases, and retrieval-augmented generation (RAG). These innovations are expected to enhance reasoning capabilities, enabling more sophisticated decision-making and automation across various domains. This paper is meant to explore the current landscape of LRMs and give insight into potential future research directions.

1. Introduction

The rapid advancement of artificial intelligence (AI) has led to the development of sophisticated models capable of performing increasingly complex cognitive tasks. While Large Language Models (LLMs) such as GPT-4 and Gemini have demonstrated remarkable proficiency in text generation, their ability to engage in structured, multi-step reasoning remains limited. To address this gap, Large Reasoning Models (LRMs) have emerged

as a novel class of AI systems specifically designed to enhance logical reasoning, structured problem-solving, and iterative inference. Unlike traditional LLMs, which primarily predict the next token in a sequence based on statistical correlations, LRMs leverage explicit reasoning techniques, including Chain-of-Thought (CoT) prompting, reinforcement learning, and structured reasoning datasets, to improve decision-making accuracy. This paper explores the motivation behind LRMs, their key training methodologies, and the implications of these advancements for real-world applications. Additionally, it examines emerging trends and future directions in LRM research, including the integration of external databases and knowledge graphs to enhance their reasoning capabilities.

2. Background and Related Work

The foundation of LRMs builds upon several key AI methodologies, including supervised learning, reinforcement learning, and retrieval-augmented generation. Traditional LLMs such as GPT-4 and Gemini rely on extensive pretraining on large-scale text corpora, fine-tuned using supervised and reinforcement learning techniques. However, these models often struggle with logical consistency and multi-step reasoning. To bridge this gap, LRMs integrate novel techniques, such as Reinforcement Learning with Human Feedback (RLHF), which optimizes reasoning steps rather than just output quality. Additionally, rejection sampling and supervised fine-tuning further enhance model performance by iteratively refining responses. OpenAI’s o1, DeepSeek’s R1, and Google’s Gemini 2.0 Flash Thinking exemplify this new paradigm, utilizing reasoning-specific training objectives to outperform conventional LLMs in complex problem-solving tasks. Furthermore, the incorporation of symbolic AI elements, external knowledge retrieval, and hybrid neural-symbolic models represents a significant step toward more interpretable and reliable AI reasoning systems.

Research on AI reasoning has gained significant momentum in recent years, with various organizations contributing to the development of more capable models. OpenAI’s o1 employs iterative problem-solving and CoT prompting to enhance logical consistency, a technique that has been widely studied in reasoning research. DeepSeek’s R1 introduces a novel reinforcement learning framework that prioritizes reasoning efficiency over mere token prediction, reducing computational costs while improving logical inference. Prior studies, such as DeepSeekMath (Shao et al., 2024), have further explored reasoning-specific fine-tuning strategies to optimize mathematical problem-solving capabilities. Additionally, studies by Xi (2024) on reverse curriculum reinforcement learning highlight alternative approaches to reasoning optimization. Surveys such as those by Sun et al. (2024) and Plaat et al. (2024) provide comprehensive overviews of reasoning models and their evolution. Furthermore, recent discussions on reasoning-oriented architectures, such as hybrid neural-symbolic models, point to promising directions for future research. By building upon these foundational works, LRMs are poised to redefine

AI’s role in structured decision-making and logical inference, enabling more sophisticated applications across multiple domains.

3. What is a Large Reasoning Model?

Large Reasoning Models, or LRMs, represent a significant shift in the field of artificial intelligence, building on the progress made by Large Language Models (LLMs) text generation and problem-solving. While LLMs excel in generating text and replicating patterns in their training data, they often fail at large multistep problems. When it comes to problem-solving and complex reasoning, these LLMs often fall behind, failing to successfully complete many problems that humans easily do. Because of this, companies and research labs have been looking for ways to improve reasoning ability in LLMs, by proxy improving their math and programming skills. This search, along with the desire to have much more understandable systems, is what led to today’s large reasoning models. LRMs are some of the newest innovations in LLM research and product releases, marking a huge step forward in logical problem-solving ability. These models exist as an extension of LLMs; a fine-tuned version that allows better step-by-step solutions. These solutions have the added benefit of helping the user understand what the model did, and where it might have gone wrong (Grootendorst, 2025).

3.1 Primary Differences Between LLMs and LRMs

One of the primary distinctions between LRMs and traditional LLMs lies in their approach to problem-solving. LLMs typically generate responses based on probabilistic associations within their training data, which can result in superficial or contextually irrelevant answers for complex queries. In contrast, LRMs are designed to emulate a more methodical reasoning process, breaking down problems into sequential steps that mirror human-like analytical thinking. This structured approach not only improves the accuracy of the solutions but also enhances the model’s ability to handle tasks that require deep logical reasoning, such as advanced mathematics, scientific research, and intricate programming challenges (Grootendorst, 2025).

Transparency and interpretability are also markedly improved in LRMs. Traditional LLMs often function as opaque systems, providing outputs without insight into the underlying reasoning pathways, which can be problematic in applications where understanding the rationale behind a decision is crucial. LRMs, however, offer detailed, step-by-step solution traces that allow users to follow the model’s logic. This feature not only aids in verifying the correctness of the output but also facilitates the identification and rectification of potential errors in the reasoning process, thereby fostering greater trust and reliability in AI-assisted decision-making (Khatib, 2025).

The development and deployment of LRMs involve a paradigm shift in computational resource allocation. While LLMs primarily concentrate computational efforts during the

training phase, called train-time-compute, with relatively lower demands during inference, LRMs require substantial computational resources during both training and inference. Focusing computation during inference, or test-time-compute, was a shift that changed the focus of these models, and caused a huge shift in resource allocation. This is due to the dynamic and iterative nature of their reasoning processes, which necessitate increased computational power to perform complex, step-by-step analyses in real-time. Consequently, implementing LRMs effectively requires careful consideration of computational infrastructure to balance the benefits of enhanced reasoning capabilities with the associated resource demands (Grootendorst, 2025).

3.2 Current Implementations

OpenAI’s o1 is designed to "think" before responding, employing techniques such as chain-of-thought prompting and iterative verification to improve logical consistency and accuracy in problem-solving tasks. This approach allows the model to break down complex problems into sequential steps, mirroring human-like analytical processes (Knight, 2024).

DeepSeek’s R1 emphasizes cost-efficient reinforcement learning to enhance reasoning capabilities. The model incorporates innovative training techniques, such as "cold start" and reasoning-oriented reinforcement learning, which prioritize reasoning efficiency over mere token prediction. This approach has enabled DeepSeek to develop powerful AI models with reduced computational costs, outperforming competitors in both performance and resource efficiency (DeepSeek-AI et al., 2025).

Google’s Gemini 2.0 Flash Thinking introduces emergent reasoning features in multimodal contexts, integrating modalities beyond text, such as vision and audio. This integration allows the model to extend its reasoning to external databases and tools, enhancing its problem-solving abilities across various domains. The incorporation of retrieval-augmented generation techniques further improves the model’s accuracy by combining internal reasoning with the retrieval of relevant external information (Thinking, 2025).

In summary, Large Reasoning Models (LRMs) represent a fundamental shift in artificial intelligence, advancing beyond the pattern-matching capabilities of traditional LLMs to enable more structured, step-by-step problem-solving. By prioritizing interpretability, logical consistency, and computational efficiency, these models offer significant improvements in fields requiring deep reasoning, such as mathematics, programming, and scientific research. The development of LRMs by leading AI organizations—such as OpenAI’s o1, DeepSeek’s R1, and Google’s Gemini 2.0 Flash Thinking—highlights the growing emphasis on AI systems that not only generate responses but also justify their reasoning in a transparent and verifiable manner. However, achieving these advanced capabilities requires innovative training methodologies that go beyond conventional deep-learning approaches. The next section explores the training processes behind LRMs, including specialized datasets, fine-tuning strategies, and reinforcement learning techniques that

enhance their reasoning abilities.

4. How Are They Trained?

The training process for LRMs builds heavily on many of the approaches that LLM training uses, though several key modifications are introduced to tailor the models’ outputs to their specific tasks. In fact, almost all current LRM training begins in the same way, with a large pretrained LLM. This base LLM serves as the foundation that will be built upon in the training process, through processes like fine tuning and reinforcement learning. For OpenAI, this base model is likely GPT 4, and for DeepSeek it is their most recent LLM, named V3 (DeepSeek-AI et al., 2025; OpenAI, 2024).

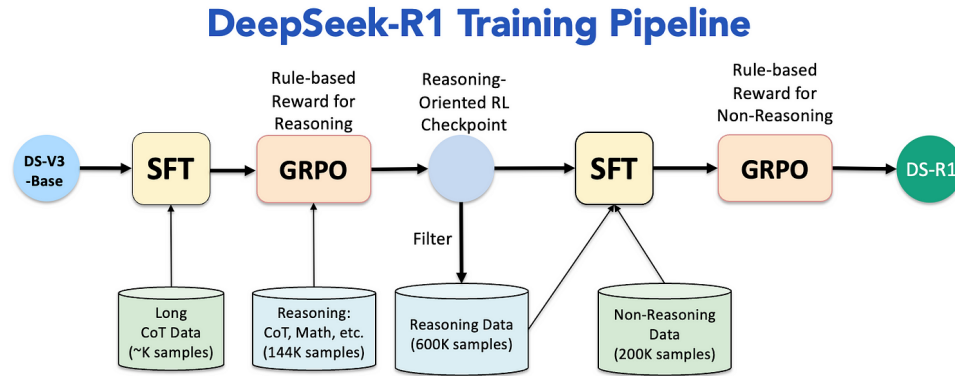


Figure 1: DeepSeek Training Pipeline. Adapted from Po (2025)

In this section, we examine the training process of DeepSeek’s R1, as illustrated in Figure 1, and provide a comparative analysis with OpenAI’s o1 model. Each section of the pipeline above will be discussed, and its contribution to the final model analyzed. There will be a higher emphasis placed on the DeepSeek model, as DeepSeek offers significantly more transparency into their model development process, including detailed documentation on both training techniques and the datasets involved. OpenAI, by contrast, is much more restrictive when it comes to details surrounding the training and data that go into their models, making it hard to dissect them beyond what they release in the model’s white paper. DeepSeek’s openness allows for a more in-depth exploration of R1’s training innovations, particularly how DeepSeek managed to develop the model at a fraction of OpenAI’s cost and with considerably fewer resources.

4.1 Pretraining Process

The first step in the development of an LRM is to assemble a high quality, reasonably sized, and diverse dataset of reasoning traces. These “reasoning traces” are examples of conversations that include thinking steps, and reasoning processes. This data is called the “cold start” data, as it is the first thing the model is shown to introduce reasoning behavior to the system.

Example Reasoning Trace

User Query:

What causes the seasons on Earth?

Thinking:

- **Identify Concepts:** Recognize that seasons relate to Earth’s tilt and orbit.
- **Recall Facts:** Earth’s axis is tilted by approximately 23.5° and it orbits the Sun annually.
- **Apply Reasoning:** The tilt causes different regions to receive varying amounts of sunlight over the year, leading to seasonal changes.

Output:

The seasons are caused by the tilt of Earth’s axis combined with its orbit around the Sun, which results in varying levels of sunlight and temperature changes throughout the year.

Figure 2: A high-quality reasoning trace illustrating the process from user query through detailed thinking to the final output.

Figure 2 shows what one of these reasoning traces might look like in this dataset, but it is hard to know for sure. While companies open source their models like DeepSeek R1, their data is not open source, so this is not an exact reasoning trace, but an example of what they would look like. Regardless, this data set is then used to fine-tune the LLM, helping the model quickly start generating reasoning traces, avoiding the unstable early phases seen in a pure reinforcement learning approach like DeepSeek R1-Zero. Despite the cold start data, much work is still needed to improve the quality and readability of the reasoning steps and output. Unlike standard language models, the pretraining objectives here are tailored to handle structured data and logical problems. The training is designed to imbue the model with the ability to work through reasoning-heavy datasets, ensuring that it can process and generate complex, multi-step responses. This is where reinforcement learning comes in, and helps these models further expand their reasoning abilities (DeepSeek-AI et al., 2025).

4.2 Training Process

4.2.1 Reinforcement Learning - Reasoning Focus

The next step in DeepSeek R1’s training pipeline is reinforcement learning (RL); more specifically, a variant called Group Relative Policy Optimization. GRPO is a new variant of the very popular RL method called proximal policy optimization (PPO). GRPO was introduced in a paper by Shao et al. (2024) at DeepSeek research in response to the high compute demands of RL methods like PPO. GRPO is focused on training these LRM models more quickly, and with less compute than before. The full mathematical

background for this reinforcement learning system is out of the scope of this paper, but the basic idea can be explained fairly simply.

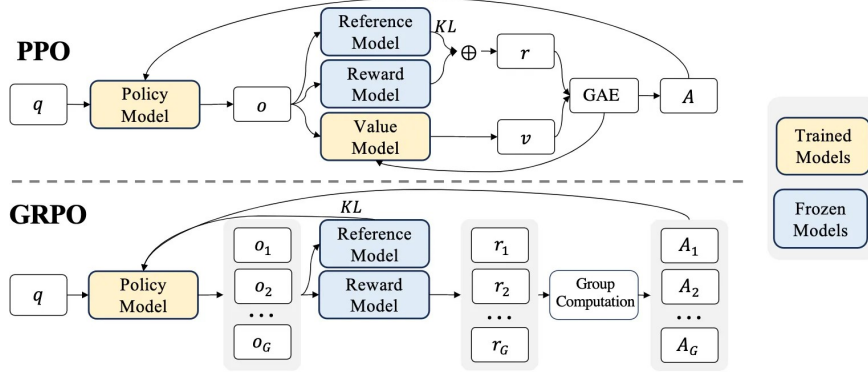


Figure 3: GRPO algorithm versus PPO algorithm, notice the lack of a value model in GRPO. Adapted from Shao et al. (2024)

Unlike PPO, which requires a large value model to estimate how good each output is, GRPO estimates quality by comparing a group of generated outputs for the same prompt and rewarding outputs that score above the group average. As shown in 3, this process eliminates the need for a separate value model, which is one of the key sources of memory and compute overhead in PPO. By simplifying the pipeline and shifting from per-token reward estimation to whole-response comparison within a group, GRPO reduces both training time and overall computational cost.

During this reinforcement learning stage, DeepSeek set out to fix a common problem that reasoning models experience in their thinking phase: language mixing. This occurs when the model switches languages as it thinks, sometimes back and forth between multiple different languages. While this isn’t always a bad thing for model output, it does take away from the end result of user interpretability, keeping users from being able to read the entire thinking trace. To combat this, DeepSeek introduced a language consistency reward, reinforcing the model to maintain a single language throughout thinking and output generation (DeepSeek-AI et al., 2025).

This approach not only reduces memory and computational overhead, but it also effectively improves the model’s reasoning ability, especially for multi-step mathematical problems. As the reinforcement training runs, the model’s outputs improve over time, meaning the average quality response improves over time as well. This strategy of selecting and rewarding the above-average responses reduces both training time and cost greatly, and was a big breakthrough that allowed DeepSeek to achieve the results that they did (Shao et al., 2024).

4.2.2 Rejection Sampling and Synthetic Data

After significant reinforcement learning had occurred and the model outputs were fairly high-quality, DeepSeek AI employed a technique known as self-generated data training.

This involves selecting the highest quality reasoning traces the model produces during reinforcement learning, and saving them in a database for later. These reasoning traces, while not perfect, still represent a fairly high quality dataset that is purely generated by the model. For companies like DeepSeek, which have limited resources compared to OpenAI or Google, this new data can be very useful (DeepSeek-AI et al., 2025).

For R1 specifically, around 600 thousand reasoning traces were selected, along with 200 thousand non reasoning answers. These reasoning traces were highly filtered from a much larger set, retaining only those with single language answers, no code blocks, and no long paragraphs. The non-reasoning answers were similarly filtered, allowing the model to take in only the highest quality generated data. In contrast to the reasoning traces, the non-reasoning answers were generated by the pure base model, DeepSeek V3 (DeepSeek-AI et al., 2025).

From there, the sampled data can be used similarly to the cold start data to fine-tune the model further. Even though this data is generated by the model itself, fine tuning on it will still yield tangible improvements to readability and consistency of reasoning. This finetuning represents the third major step in the DeepSeek R1 training pipeline, and the second to last step overall (DeepSeek-AI et al., 2025).

4.2.3 Reinforcement Learning - Non-Reasoning Focus

The final training step of DeepSeek R1 involves a non-reasoning focused reinforcement learning stage. This reinforcement learning has the key goal of improving the final output of the model, putting very little focus on editing or evaluating the thinking output. This improves the simplicity of the final stage, and allows DeepSeek to tune the final output how they like, focusing on readability and helpfulness. While not specific in how the reward system improves model output, DeepSeek AI mentions that

“For reasoning data, we adhere to the methodology outlined in DeepSeek-R1-Zero, which utilizes rule-based rewards to guide the learning process in math, code, and logical reasoning domains.”

DeepSeek-AI et al., 2025

These rule-based rewards serve as the final step, and help the model improve its usability and understandability. After this step is complete, the model is mostly ready to ship to the user, besides training smaller versions of the model through steps such as distillation.

4.3 Distillation Process in DeepSeek

Distillation is a pivotal technique in DeepSeek’s model optimization strategy, aimed at transferring knowledge from larger, more capable models to smaller, efficient ones without significant performance loss. This process involves training a compact student model to

replicate the behavior of a larger teacher model, thereby reducing computational requirements while maintaining high levels of accuracy and reasoning ability (DeepSeek-AI et al., 2025).

In the development of DeepSeek-R1, the distillation process began by synthesizing a substantial dataset using the internal model. As stated above, 600,000 reasoning traces were generated, focusing on logical inference and problem-solving tasks. These traces were meticulously filtered to ensure quality, including checks for language consistency and the exclusion of code blocks or lengthy paragraphs. Additionally, 200,000 non-reasoning samples were produced using DeepSeek-V3, covering areas such as writing, factual question answering, self-cognition, and translation (DeepSeek-AI et al., 2025).

The student models were then fine-tuned on this curated dataset, undergoing supervised fine-tuning for two epochs. This step enabled the smaller models to learn from the high-quality outputs of the larger model, effectively capturing its reasoning patterns and linguistic nuances. Notably, these distilled models were initialized from other pre-trained open-weight models, including LLaMA and Qwen, before being fine-tuned on the synthetic data generated by DeepSeek R1 (DeepSeek-AI et al., 2025).

This distillation approach employed by DeepSeek allowed them to produce models that maintained similar performance while shrinking storage and compute requirements significantly. By leveraging synthetic data generation and targeted fine-tuning, DeepSeek successfully developed smaller models that retain robust reasoning capabilities, making them more accessible for deployment in various applications.

4.3.1 OpenAI vs. DeepSeek

While the DeepSeek R1 training process is fairly representative of trends within the field, OpenAI (and other companies) have their own proprietary techniques that they keep as hidden as possible from the public. While not everything is known about OpenAI o1 and o3 training, there are some things that we do know, and other things we can speculate about. One of the biggest differences in OpenAI’s training, when compared to DeepSeek, is in the heavy use of reinforcement learning with human feedback, or RLHF. This type of training involves a large number of people with the sole task of grading and evaluating the model responses and thinking traces. This technique has the advantage of allowing OpenAI and the graders to adjust the model rewards based on human values such as readability and flow. These concepts are easy for a human to understand, but hard to evaluate programmatically. This means that a pure RL technique like GRPO will have a hard time competing with RLHF in some tasks and areas, specifically related to understandability and consistency (OpenAI, 2024; Xu et al., 2025).

On the other hand, RLHF is extremely costly both monetarily and time-wise when compared to GRPO or other pure RL systems. This means that only companies with huge budgets and training times such as OpenAI or Google can employ this strategy, leaving DeepSeek and other smaller companies to optimize their pure RL systems. As

a result of this, DeepSeek got really good at RL, leading to their huge push forward in making cheaper and more efficient models(OpenAI, 2024; Xu et al., 2025).

4.4 DeepSeek’s Impact on AI Hardware Market

The evolving demands of artificial intelligence, particularly in inference computing, are reshaping the AI hardware landscape. Inference computing, which involves running trained models to make predictions or decisions, now requires increased computational power as user interactions with AI models become more complex and frequent. This shift has opened opportunities for companies to challenge established players like Nvidia by developing specialized chips optimized for inference tasks (Heikkilä, 2025).

DeepSeek has emerged as a significant contributor to this trend, advancing models that excel in inference efficiency. Their innovations have prompted a broader industry movement toward developing hardware tailored for inference workloads. This includes efforts by startups such as Cerebras and Groq, as well as tech giants like Google, Amazon, Microsoft, and Meta, all aiming to create more efficient alternatives to traditional GPUs for AI inference (Heikkilä, 2025).

Analysts predict a substantial increase in investments for inference infrastructure, anticipating that inference will soon constitute the majority of AI-related computational demand. While Nvidia’s latest Blackwell chips are designed to perform well in inference scenarios, the emergence of more efficient, specialized hardware solutions could significantly alter the AI chip market’s dynamics by 2028 (Heikkilä, 2025).

DeepSeek’s contributions to AI model efficiency and the resulting shifts in hardware requirements highlight the interconnected nature of software advancements and hardware development. As AI models become more sophisticated and widely used, the demand for optimized inference hardware is likely to continue growing, fostering innovation and competition in the AI hardware sector (Heikkilä, 2025).

5. What is the Future of LRMs?

When it comes to the future of LRMs, it is hard to say exactly what direction the field will go in. Although there are general trends in the industry, this section will mainly be speculation and opinion. Knowing the future for these models is impossible, but we hope that, based on the trends evolving currently, a general trend can be identified.

5.1 Formal Logic - Symbolic AI

One of the big issues currently facing LRMs and research teams developing them is the logical consistency of the answers and correctness. While these are huge problems to solve, something that has been left out of most commercial models (to the public’s knowledge) is a formal logic system. A formal logic system is a way of representing answers and

information that allows the model to maintain consistency throughout its thinking and output. In Lyu et al. (2023) the authors propose a formal logic chain of thought system that would allow models to symbolically represent their arrival at an answer. A common problem with current chains of thought is that they can "lie" about how they really came to an answer. This system claims to be able to mitigate this, and allow models to improve their accuracy in the process.

If a system like this could be scaled to the size of OpenAI and DeepSeek's models, it could revolutionize the way that models represent chains of thought. A symbolic approach could allow models to be both more accurate than ever, but also more interpretable. While the scalability and success of a system like this is still unknown, it would be surprising if labs were not trying to work this into their systems (Lyu et al., 2023).

5.2 External Agent Integration

For a while, a huge drawback of major LLMs and reasoning models was their lack of access to the newest data and internet resources. Until recently, LLMs could help summarize research, but there was little they could do to help you find sources or articles related to your prompts. This changed with the release of ChatGPT internet integration and Deep Research (by multiple companies). These systems will now scour the web for sources related to a question if you ask, allowing you to find niche articles or papers related to whatever field you are interested in. Especially Deep Research, which will often take upwards of 30 minutes to run its course, claims to conduct a very in-depth search of the web (OpenAI, 2025).

While this challenge of external knowledge integration isn't solved, and will surely be improved over the next few years, these tools are a big first step. Leveraging LRMs and LLMs to find information, rather than just regurgitate training data is a huge plus, and makes them even more useful to students and professionals alike. The future of these systems is likely built on tools like Deep Research, allowing LRMs greater access to the outside internet, and potentially even the outside world. Agentic design similar to deep research could allow LRMs to perform almost any task on the internet, assuming they have the training for it.

6. Conclusion

This paper has explored the growing field of LRMs, highlighting their distinct advantages over traditional Large Language Models (LLMs) in complex problem-solving and logical inference. By examining key examples such as OpenAI's o1, DeepSeek's R1, and Google's Gemini 2.0 Flash Thinking, we have demonstrated the innovative training methodologies that enable LRMs to excel in tasks requiring deep reasoning. These methodologies, including structured reasoning datasets, Reinforcement Learning with Human Feedback (RLHF), and iterative problem-solving, represent a significant advancement in AI capab-

ilities. DeepSeek’s approach, particularly its focus on cost-efficient reinforcement learning and the use of synthetic data, exemplifies the potential for developing high-performance LRMs with fewer resources.

Looking forward, the future of LRMs appears promising, with potential advancements in formal logic integration and external agent integration. The incorporation of symbolic AI elements and knowledge graphs could enhance logical consistency and interpretability, addressing current limitations in reasoning accuracy. By continuing to innovate and refine training methodologies, LRMs are poised to redefine the landscape of artificial intelligence, offering unprecedented capabilities in logical reasoning and structured problem-solving.

References

- DeepSeek-AI et al. (2025). Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv*, (arXiv:2501.12948). <https://doi.org/10.48550/arXiv.2501.12948>
- Grootendorst, M. (2025). A visual guide to reasoning llms [Exploring Language Models]. <https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-reasoning-llms>
- Heikkilä, M. (2025). Deepseek’s “aha moment” creates new way to build powerful ai with less money [FinancialTimes; Financial Times]. <https://www.ft.com/content/ea803121-196f-4c61-ab70-93b38043836e>
- Khatib, M. (2025). The era of reasoning models — moving beyond chain of thought prompting [Medium]. <https://medium.com/@mayadakhatib/the-era-of-reasoning-models-moving-beyond-chain-of-thought-prompting-fa24ce25c106>
- Knight, W. (2024). Openai announces a new ai model, code-named strawberry, that solves difficult problems step by step. *WIRED*. <https://www.wired.com/story/openai-o1-strawberry-problem-reasoning/>
- Lyu, Q., Havaldar, S., Stein, A., Zhang, L., Rao, D., Wong, E., Apidianaki, M., & Callison-Burch, C. (2023). Faithful chain-of-thought reasoning. *ArXiv.org*. <https://arxiv.org/pdf/2301.13379>
- OpenAI. (2024). Learning to reason with llms [openai.com/index/learning-to-reason-with-llms]. <https://openai.com/index/learning-to-reason-with-llms>
- OpenAI. (2025). Introducing deep research [Accessed: 2025-03-06]. <https://openai.com/index/introducing-deep-research/>
- Plaat, A., Wong, A., Verberne, S., Broekens, J., Stein, N. v., & Back, T. (2024). Reasoning with large language models, a survey. *arXiv*, (arXiv:2407.11511). <https://doi.org/10.48550/arXiv.2407.11511>
- Po, L. (2025). Deepseek-r1: Affordable, efficient, and state-of-the-art ai reasoning [Medium]. <https://medium.com/@lmpo/deepseek-r1-affordable-efficient-and-state-of-the-art-ai-reasoning-f293b0bd8d65>

- Shao, Z., Wang, Q., P. and Zhu, Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., K, L. Y., Wu, Y., & Guo, D. (2024). Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv*, (arXiv:2402.03300). <https://doi.org/10.48550/arXiv.2402.03300>
- Sun, J., Zheng, C., Xie, E., Liu, Z., Chu, R., Qiu, J., Xu, J., Ding, M., Li, H., Geng, M., Wu, Y., Wang, W., Chen, J., Yin, Z., Ren, X., Fu, J., He, J., Yuan, W., Liu, Q., & Li, Z. (2024). A survey of reasoning with foundation models. *arXiv*, (arXiv:2312.11562). <https://doi.org/10.48550/arXiv.2312.11562>
- Thinking, G. 2. F. (2025). Google cloud. <https://cloud.google.com/vertex-ai/generative-ai/docs/thinking>
- Xi, Z. e. a. (2024). Training large language models for reasoning through reverse curriculum reinforcement learning. *arXiv*. <https://doi.org/10.48550/arXiv.2402.05808>
- Xu, F., Hao, Q., Zong, Z., Wang, J., Zhang, Y., Wang, J., Lan, X., Gong, J., Ouyang, T., Meng, F., Shao, C., Yan, Y., Yang, Q., Song, Y., Ren, S., Hu, X., Li, Y., Feng, J., Gao, C., & Li, Y. (2025). Towards large reasoning models: A survey of reinforced reasoning with large language models. *ArXiv.org*. <https://arxiv.org/abs/2501.09686>